Module 2

Single phase Fully controlled and Half controlled Converters

SINGLE-PHASE FULL-WAVE BRIDGE CONVERTERS

- Phase controlled 1-phase or 3-phase, full converters – 3 types

  - uncontrolled converters, half-controlled & fully-controlled

- Uncontrolled converter – uses only diodes, level of output voltage cannot be controlled
- Half-controlled converter or semiconverter – uses a mixture of diodes & thyristors, limited control over the level of DC output voltage
- Fully-controlled converter or full converter –uses thyristors only, wider control over the level of DC output voltage

SINGLE-PHASE FULL-WAVE BRIDGE CONVERTERS

- Semiconverter is one-quadrant converter
- One-quadrant converter – one polarity of DC output and current at its output
- Two-quadrant converter – voltage polarity can reverse but current direction cannot reverse because of the unidirectional nature of thyristors.

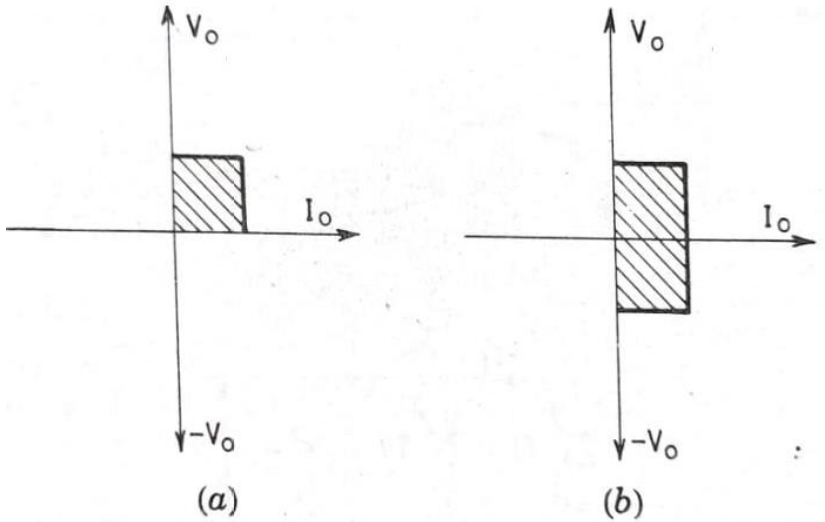
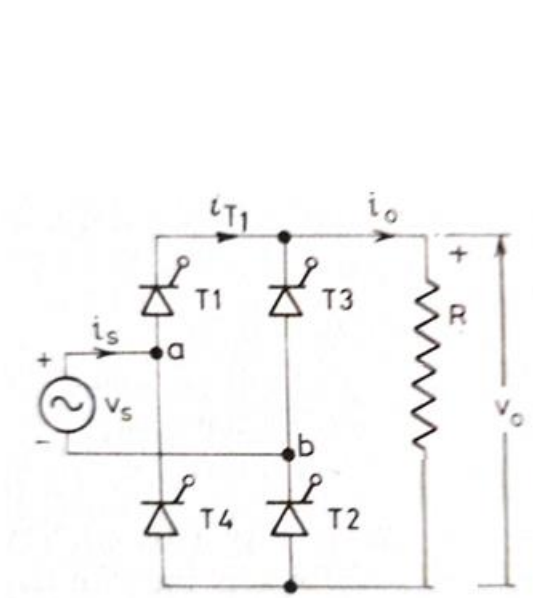
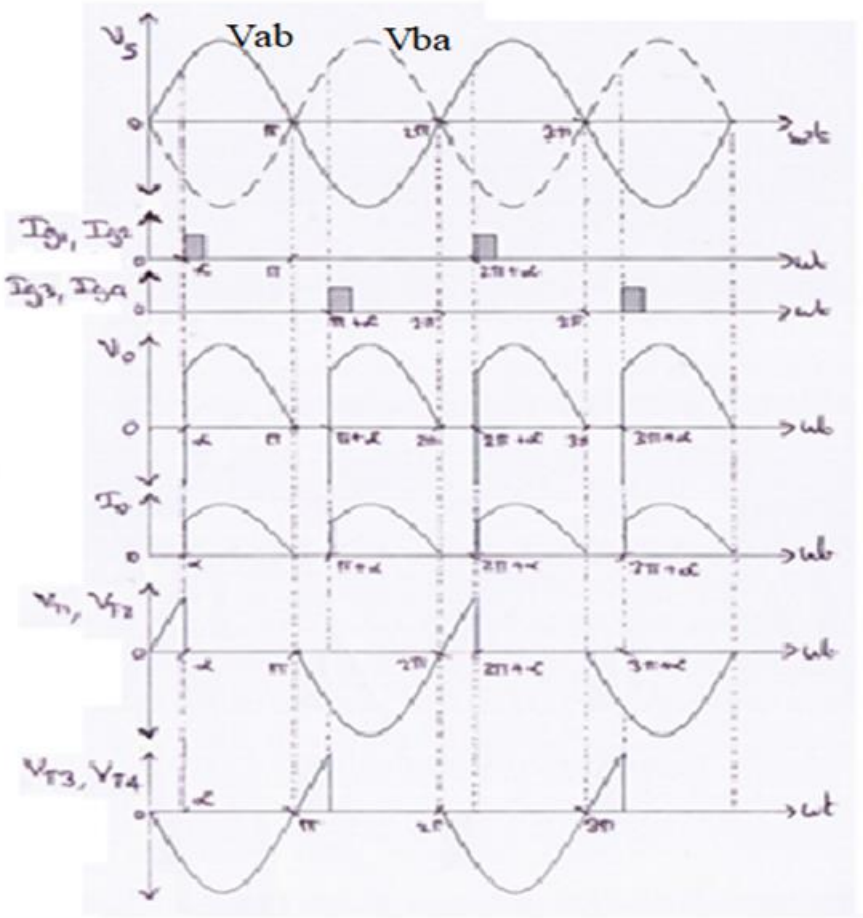


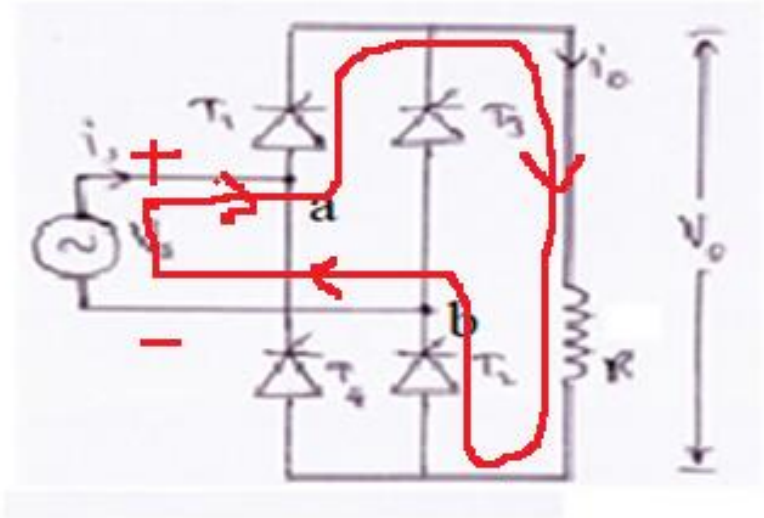
Fig. (a) One-quadrant converter and (b) two-quadrant converter.

SINGLE PHASE FULLY CONTROLLED BRIDGE RECTIFIER-R LOAD

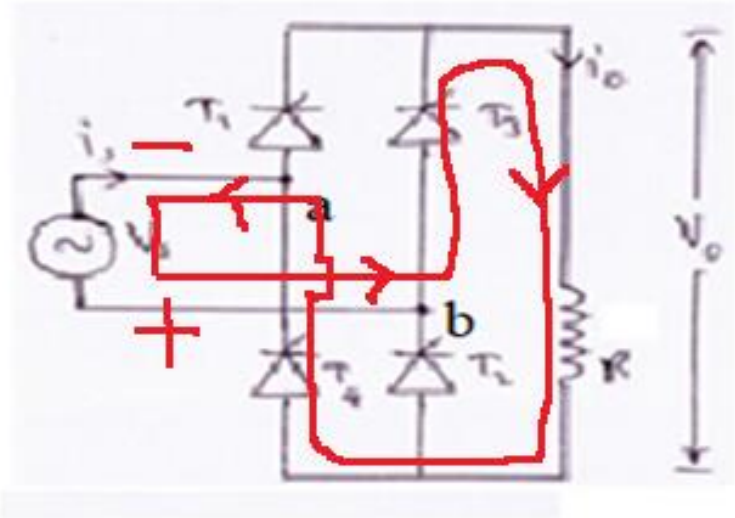




SINGLE PHASE FULLY CONTROLLED BRIDGE RECTIFIER-R LOAD



+ve half cycle



-ve half cycle

SINGLE PHASE FULLY CONTROLLED BRIDGE RECTIFIER-R LOAD

- From $\omega t=0$ to $\omega t=\alpha$

  $\mathbf{v_o=0, i_o=0}$, v_{T1} & $v_{T2} = v_s$ and v_{T3} & $v_{T4} = -v_s$

- From $\omega t= \alpha$ to $\omega t=\pi$

  $\mathbf{v_o= v_s, i_o= v_s/R}$, v_{T1} & $v_{T2} = 0$ and v_{T3} & $v_{T4} = -v_s$

- From $\omega t= \pi$ to $\omega t= \pi+\alpha$

  $\mathbf{v_o=0, i_o=0}$, v_{T1} & $v_{T2} = -v_s$ and v_{T3} & $v_{T4} = v_s$

- From $\omega t= \pi+\alpha$ to $\omega t=2\pi$

  $\mathbf{v_o= v_s, i_o= v_s/R}$, v_{T1} & $v_{T2} = -v_s$ and v_{T3} & $v_{T4} = 0$.

$$V_0 = \frac{1}{\pi}\int_{\alpha}^{\pi} V_m \sin \omega t \, . \, d(\omega t) = \frac{V_m}{\pi} \, (1 + \cos \alpha)$$

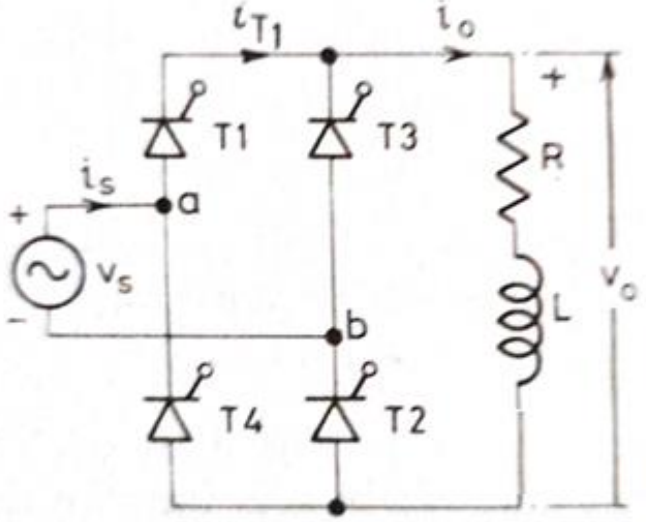
rms value of output voltage is

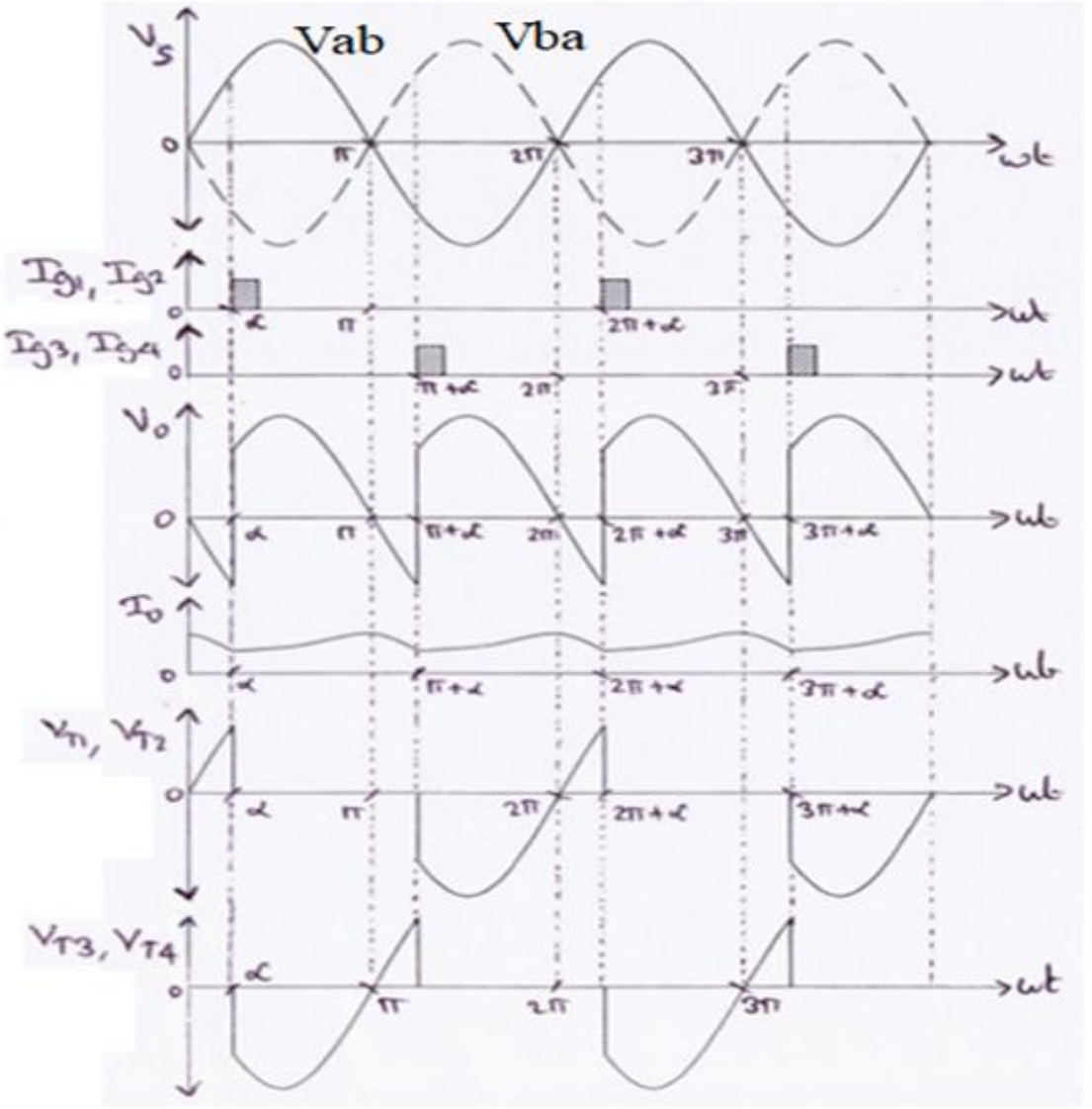
$$V_{or}^2 = \frac{1}{\pi}\int_{\alpha}^{\pi} V_m^2 \sin^2 \omega t \, d(\omega t)$$

$$= \frac{V_m^2}{2\pi}\left[\left| \omega t - \frac{\sin 2\omega t}{2} \right|_{\alpha}^{\pi} \right] = \frac{V_s^2}{\pi}\left[(\pi - \alpha) + \frac{\sin 2\alpha}{2} \right]$$

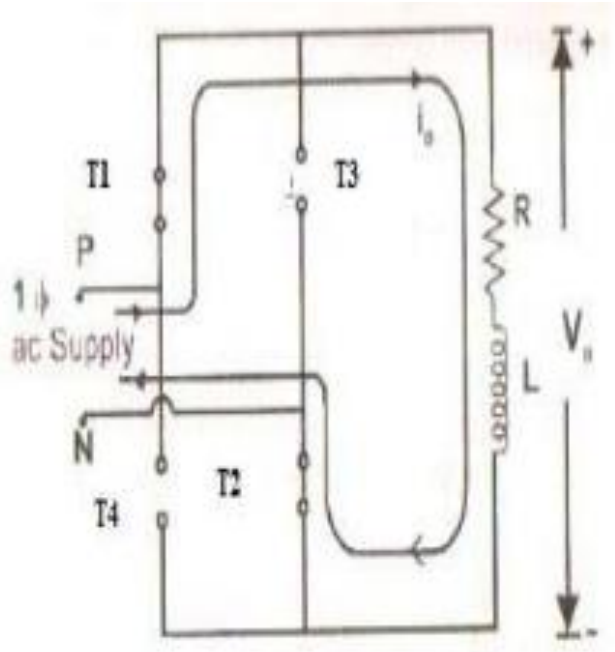
$$\therefore \quad V_{or} = V_s \left[\frac{1}{\pi} \left\{ (\pi - \alpha) + \frac{\sin 2\alpha}{2} \right\} \right]^{1/2}$$

SINGLE PHASE FULLY CONTROLLED BRIDGE RECTIFIER-RL LOAD

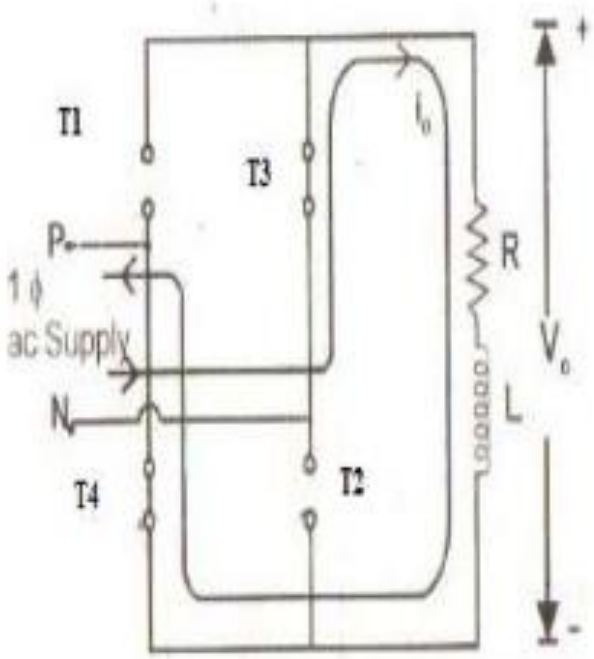




SINGLE PHASE FULLY CONTROLLED BRIDGE RECTIFIER-RL LOAD



during +ve half cycle



during -ve half cycle

SINGLE PHASE FULLY CONTROLLED BRIDGE RECTIFIER-RL LOAD

- From $\omega t = \alpha$ to $\omega t = \pi$

  $v_o = v_s$, v_{T1} & $v_{T2} = 0$ and v_{T3} & $v_{T4} = -v_s$

- From $\omega t = \pi$ to $\omega t = \pi+\alpha$

  $v_o = -v_s$, v_{T1} & $v_{T2} = 0$ and v_{T3} & $v_{T4} = v_s$

- From $\omega t = \pi+\alpha$ to $\omega t = 2\pi$

  $v_o = v_s$, v_{T1} & $v_{T2} = -v_s$ and v_{T3} & $v_{T4} = 0$.

- From $\omega t = 2\pi$ to $\omega t = 2\pi+\alpha$(also for $\omega t = 0$ to $\omega t = \alpha$)

  $v_o = -v_s$, v_{T1} & $v_{T2} = v_s$ and v_{T3} & $v_{T4} = 0$

$$V_o = \frac{1}{\pi}\left[\int_{\alpha}^{\beta} V_m \sin \omega t \, d(\omega t) + E(\pi + \alpha - \beta)\right]$$

$$= \frac{V_m}{\pi}(\cos \alpha - \cos \beta) + E\left(1 - \frac{\gamma}{\pi}\right)$$

$\gamma = \beta - \alpha$ = conduction angle

Average load current, $I_o = \frac{V_o}{R} = \frac{V_m}{\pi R}(\cos \alpha - \cos \beta) + \frac{E}{R}\left(1 - \frac{\gamma}{\pi}\right)$.

SINGLE PHASE FULLY CONTROLLED BRIDGE RECTIFIER

- By controlling the phase angle of the firing pulses in the range of 0-180 degrees, the average value of the output voltage can be varied.
- The average output voltage can be varied from +ve maximum to –ve maximum.
- The average output voltage is reversible, but the current flow in the DC terminals is unidirectional.
- i.e., the power flow can be in either direction.
- Two modes of operation is possible with fully controlled single phase bridge circuit – **rectifying mode and inverting mode**.

SINGLE PHASE FULLY CONTROLLED BRIDGE RECTIFIER

Mode 1 – Rectifying mode:-

- During α to π, both the supply voltage & supply current are +ve, so power flows from source to load.
- From π to $\pi+\alpha$, the supply voltage is –ve but supply current is +ve, so load returns power to the source.
- Since $(\pi-\alpha) > \alpha$, the net power flows from source to load.[$(\pi-\alpha)$ is the conduction period of T_1 & T_2]
- If $\alpha > 90^o$, the voltage at the output is +ve, the power flows from the source to the load and the converter operates as a rectifier.

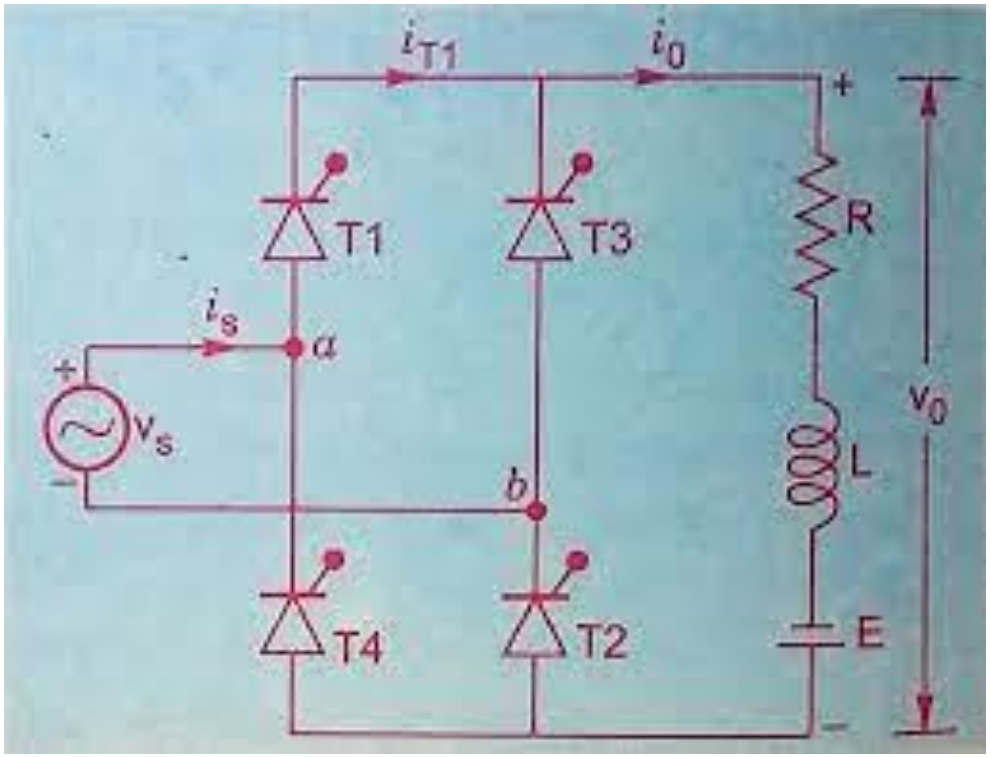
SINGLE PHASE FULLY CONTROLLED BRIDGE RECTIFIER

Mode 2 – Inverting mode:-

- When the α is greater than 90°, the dc output voltage and thus the dc power is –ve.
- This is known as the inverter operation of the converter.
- A full converter with firing angle greater than 90° is called a ***line commutated inverter***.
- The line current is displaced from the supply voltage by an angle greater than 90°, it indicates that the power is being delivered to the source.

LINE COMMUTATED INVERTER

- To achieve this in practice, a source of DC voltage which is equal to the average DC voltage must be connected at the output.
- This external source voltage (E) drives the direct current into the converter against the counter voltage produced at the output terminals.



LINE COMMUTATED INVERTER

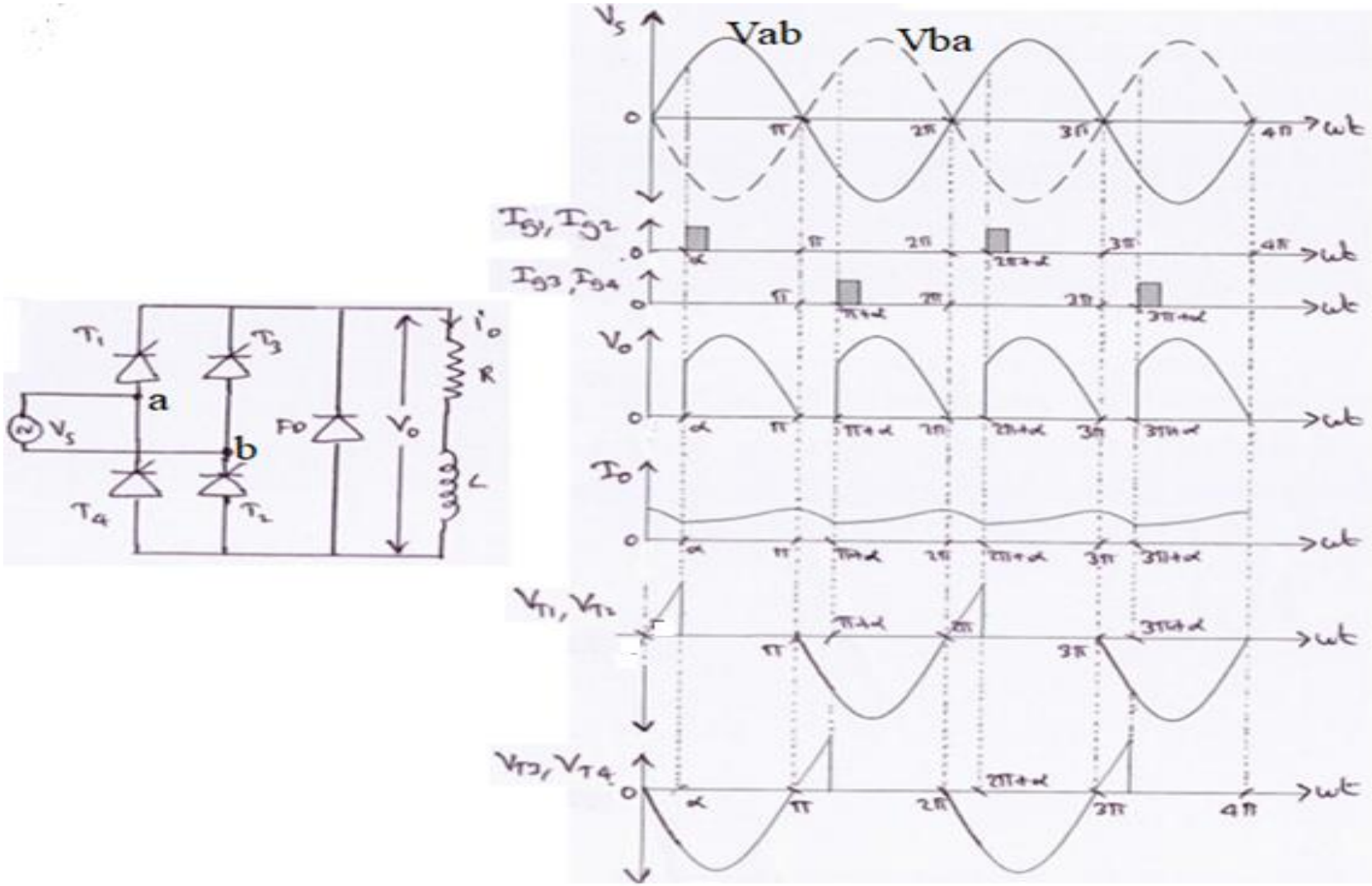
- If E is not connected, the conduction will stop somewhere before $(180^{o}+\alpha)$.
- This operation is used in the regenerative braking mode of dc drives
- In the rectifier operation, the average output voltage must be greater than the load circuit emf E.
- In the inverter operation, the load circuit emf E must be greater than the ac supply voltage.
- i.e., dc source voltage E must be more than inverter voltage Vo, then only power would flow from dc source to ac source.

LINE COMMUTATED INVERTER

- Both in rectifier and inverter modes, thyristors must be forward biased and current through SCRs must flow in the same direction as SCRs are unidirectional devices.
- Circuit turn-off time for rectifier & inverter modes are

$$t_c = \frac{\pi - \alpha}{\omega} \text{sec}$$

SINGLE PHASE FULLY CONTROLLED BRIDGE RECTIFIER- RL LOAD & FWD



SINGLE PHASE FULLY CONTROLLED BRIDGE RECTIFIER-RL LOAD & FWD

- The circuit consists of 4 thyristors T_1,T_2,T_3,T_4, a voltage source V_s ,an RL load and a freewheeling diode across the load.
- In the +ve half cycle of input cycle, T_1&T_2 are forward biased and it does not conduct until it is triggered at $\omega t=\alpha$.
- At $\omega t=\alpha$ a gate pulse is given to T_1&T_2 and it begins to conduct.
- When T_1&T_2 is ON, the input voltage is applied to the load. Due to the inductance in the load, the load current i_o increases slowly through V_s–T_1-load- T_2 - V_s
- From $\omega t=\alpha$ to $\omega t=\pi$, T_1&T_2 is ON and $v_o=v_s$ and v_{T1} &v_{T2} = 0
- During –ve half cycle, i.e. at $\omega t=\pi$, T_3&T_4 are forward biased & T_1&T_2 gets reverse biased but the current through it is not zero due to load inductor L and it will not turn off.

SINGLE PHASE FULLY CONTROLLED BRIDGE RECTIFIER-RL LOAD & FWD

- The current shifts to FD and the current circulates through FD-R-L-FD and T_1&T_2 turns off at $\omega t=\pi$.
- At $\omega t=\pi+\alpha$, T_3&T_4 is triggered and begins to conduct.
- When T_3&T_4 is ON, the load current i_o increases slowly through V_s– T_3-load- T_4 – V_s
- At $\omega t=2\pi$, T_1&T_2 is forward biased and T_3&T_4 is reverse biased. The current shifts to FD and the current circulates through FD-R-L-FD and T_3&T_4 turns off at $\omega t=2\pi$.

SINGLE PHASE FULLY CONTROLLED BRIDGE RECTIFIER-RL LOAD & FWD

- From $\omega t = \alpha$ to $\omega t = \pi$

  $v_o = v_s$, v_{T1} & $v_{T2} = 0$ and v_{T3} & $v_{T4} = -v_s$

- From $\omega t = \pi$ to $\omega t = \pi+\alpha$

  $v_o = 0$, v_{T1} & $v_{T2} = -v_s$ and v_{T3} & $v_{T4} = v_s$

- From $\omega t = \pi+\alpha$ to $\omega t = 2\pi$

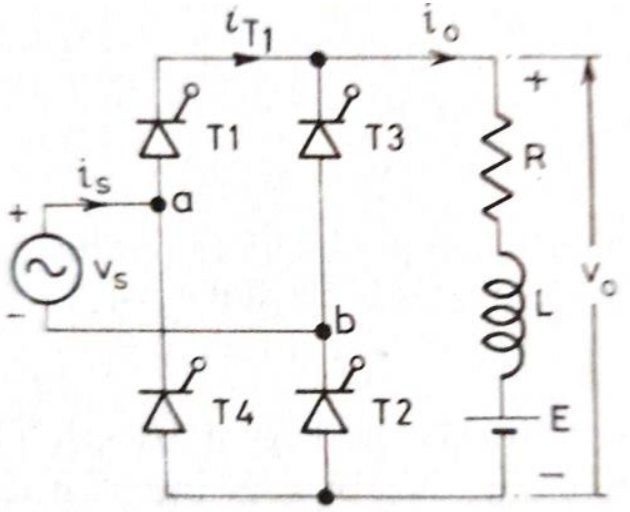
  $v_o = v_s$, v_{T1} & $v_{T2} = -v_s$ and v_{T3} & $v_{T4} = 0$.

- From $\omega t = 2\pi$ to $\omega t = 2\pi+\alpha$(also for $\omega t = 0$ to $\omega t = \alpha$)

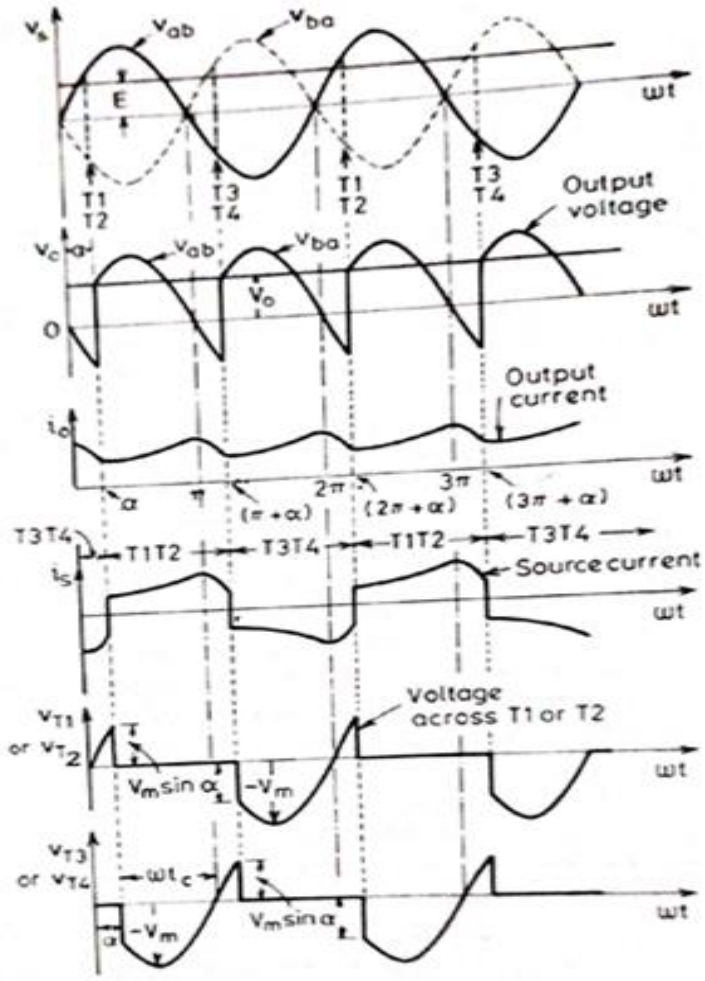
  $v_o = 0$, v_{T1} & $v_{T2} = v_s$ and v_{T3} & $v_{T4} = -v_s$

SINGLE PHASE FULLY CONTROLLED BRIDGE RECTIFIER- RLE LOAD

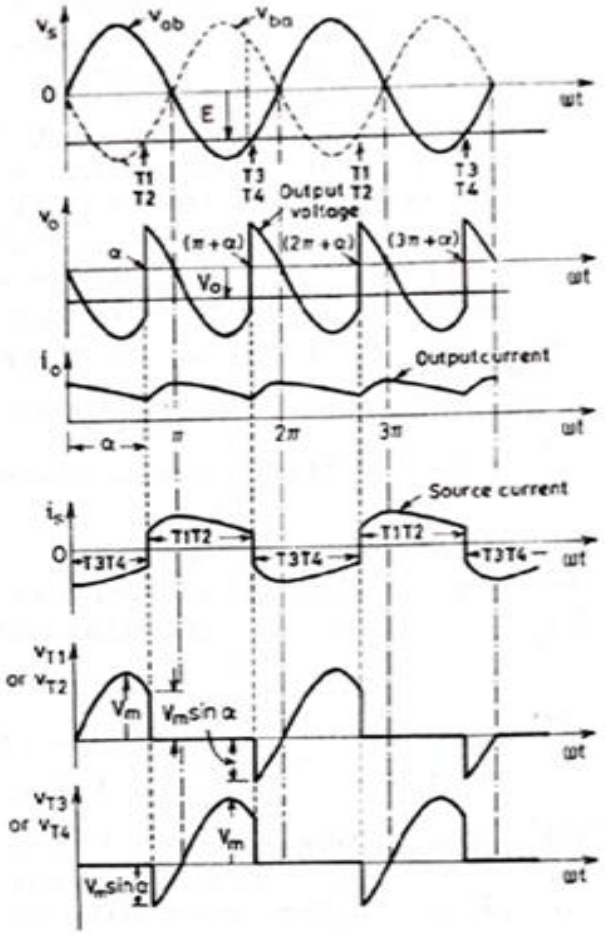
- The circuit consists of 4 thyristors T_1,T_2,T_3,T_4, a voltage source V_s ,an RLE load
- RLE load comprises of a series combination of R & L in series with a battery E.
- The emf E in the load may be a battery or the back emf of a DC motor.
- When point a is positive with respect to b, the supply voltage waveform is v_{ab}
- When point b is positive with respect to a, the supply voltage waveform is v_{ba}
- $v_{ab} = -v_{ba}$
- The load current is assumed to be continuous. i.e. the load is always connected to the AC source through the thyristors.



SINGLE PHASE FULLY CONTROLLED BRIDGE RECTIFIER-RLE LOAD



firing angle <90



firing angle >90

SINGLE PHASE FULLY CONTROLLED BRIDGE RECTIFIER-RLE LOAD

For firing angle (α)<90°:-

- From $\omega t=0$ to $\omega t=\alpha$, T_1&T_2 are forward biased through the conducting SCRs T_3&T_4.
- For continuous conduction of load current T_3&T_4 conduct after $\omega t=0$ even though they are reverse biased,
- At $\omega t=\alpha$, forward biased SCRs T_1&T_2 are triggered and they get turned on. Now the supply voltage appear across T_3&T_4 as a reverse voltage and they are turned off by natural commutation. At this instant the load current flowing through T_3&T_4 is transferred to T_1&T_2

SINGLE PHASE FULLY CONTROLLED BRIDGE RECTIFIER-RLE LOAD

- At $\omega t=\alpha$, when T_1&T_2 are gated these two will get turned on only if $V_m \sin\alpha > E$.
- T_1&T_2 will conduct from $\omega t=\alpha$ to $\omega t= \pi+\alpha$.
- At $\omega t= \pi$, T_3&T_4 are forward biased.
- At $\omega t= \pi+\alpha$, T_3&T_4 are triggered.
- Supply voltage turns off T_1&T_2 by natural commutation and i_o is transferred from T_1&T_2 to T_3&T_4.
- Maximum reverse voltage across T_1, T_2, T_3&T_4 is V_m .
- At the instant of triggering, each SCR is subjected to a reverse voltage of $V_m \sin\alpha$.

SINGLE PHASE FULLY CONTROLLED BRIDGE RECTIFIER-RLE LOAD

- Source current i_s is assumed to be +ve when T_1&T_2 are conducting and –ve T_3&T_4 are conducting.
- From $\omega t = \alpha$ to $\omega t = \pi$, v_s & i_s are +ve(because T_1&T_2 are conducting), power flows from the AC source to DC source(load).
- From $\omega t = \pi$ to $\omega t = \pi+\alpha$, v_s is –ve & i_s is +ve(because T_1&T_2 are conducting), load returns energy to the supply.
- Since $(\pi-\alpha)>\alpha$, the net power flow is from AC source to DC source.

The average value of output voltage V_0 is given by

$$V_0 = \frac{1}{\pi}\int_{\alpha}^{\pi+\alpha} V_m \sin \omega t \cdot d(\omega t) = \frac{2\,V_m}{\pi}\cos\alpha$$

- From this equation, if $\alpha > 90^0$ then V_0 is –ve.

SINGLE PHASE FULLY CONTROLLED BRIDGE RECTIFIER-RLE LOAD

For firing angle (α)>90°:-

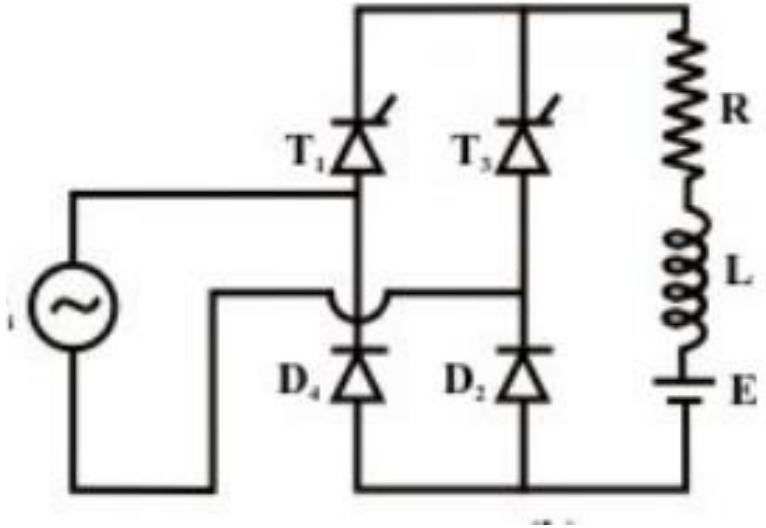
- If load circuit emf E is reversed, E will feed power back to the supply. This operation of full converter is known as the inverter operation of the converter.
- A full converter with $\alpha > 90^0$ is called line commutated inverter.
- This operation is used in regenerative braking mode of a DC motor. where E is the back emf of the DC motor.
- From $\omega t = 0$ to $\omega t = \alpha$ v_s is +ve & i_s is -ve(because T_3&T_4 are conducting), power flows from the DC source to AC source.
- From $\omega t = \alpha$ to $\omega t = \pi$, v_s & i_s are +ve(because T_1&T_2 are conducting), power flows from the AC source to DC source.
- Since $(\pi-\alpha)<\alpha$, the net power flow is from DC source(load) to AC source.

SINGLE PHASE FULLY CONTROLLED BRIDGE RECTIFIER-RLE LOAD

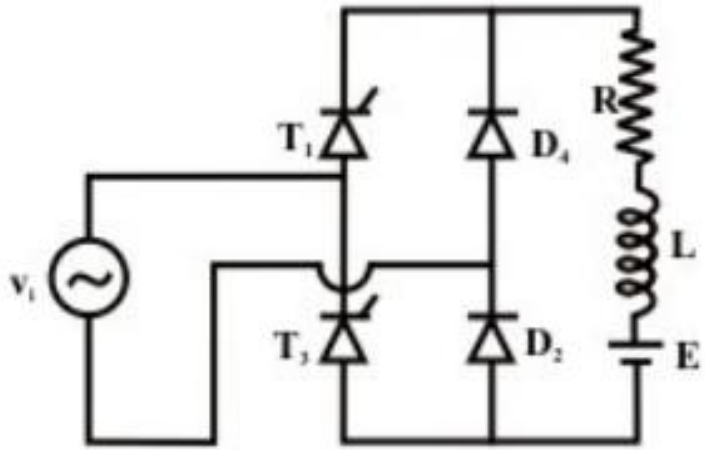
- In converter operation, average output voltage V_o is > E.
- During inverter operation, E when inverted must be greater than AC supply voltage or E must be greater than inverted voltage V_o only then power would flow from DC source to AC source.
- In both inverter & converter modes, thyristors must be forward biased and the current through SCRs must flow in the same direction as these are unidirectional devices. That's why output current is +ve
- Circuit turn-off time, $t_c = (\pi-\alpha)/\omega$ seconds, where $\omega= 2\pi f$, f=supply frequency in Hz.

Half controlled Converter

- Semi converter
- One quadrant converter
- With R load - Same as Full converter

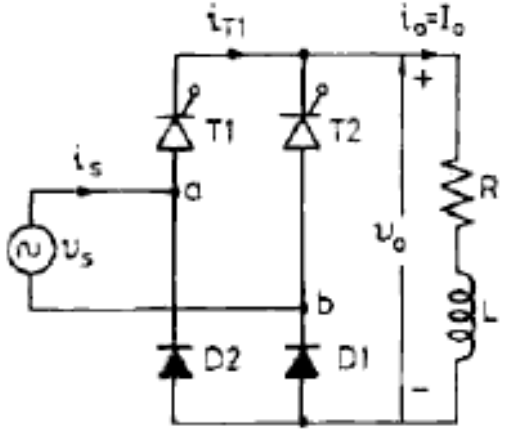


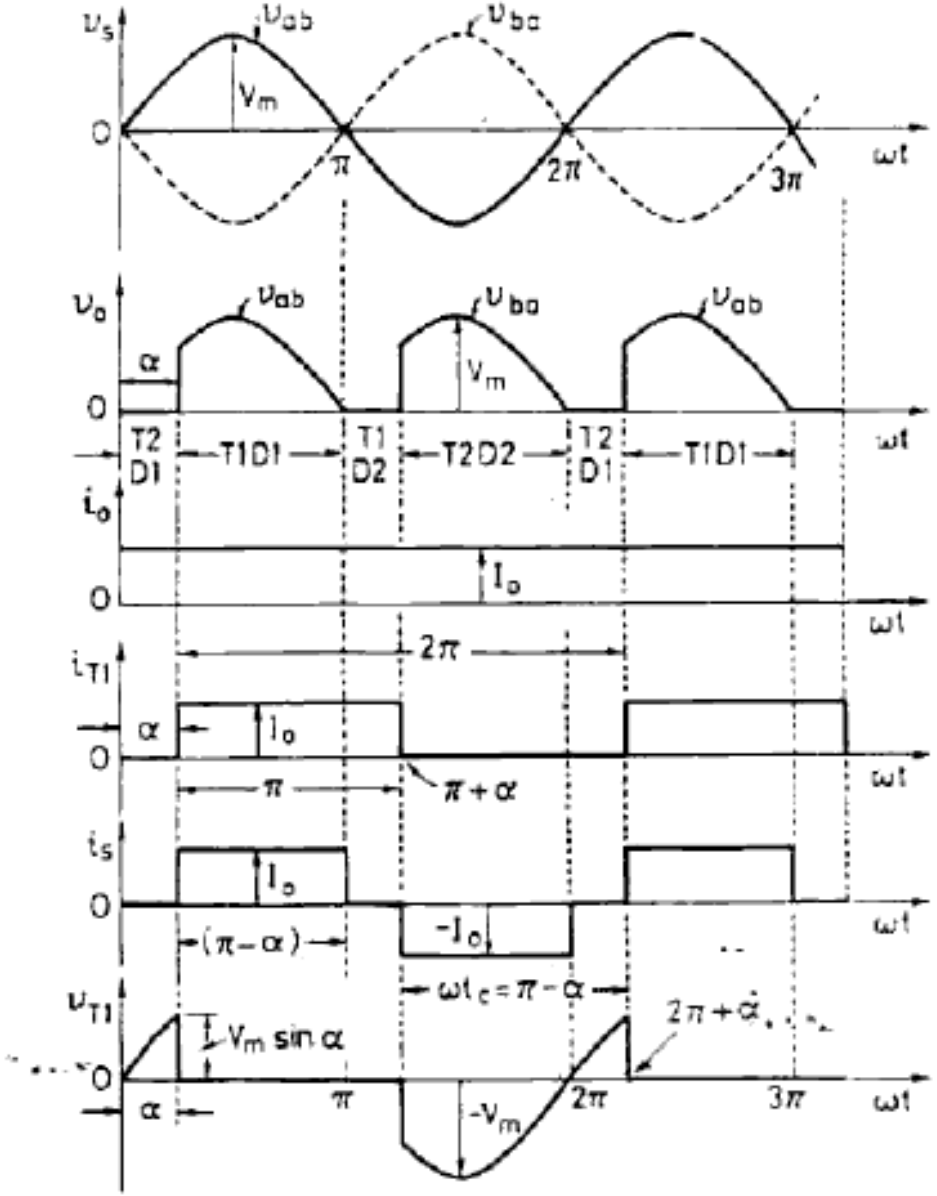
Symmetrical



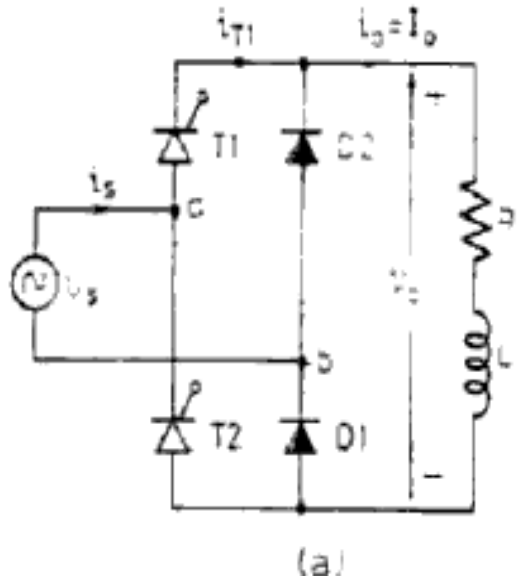
Asymmetrical

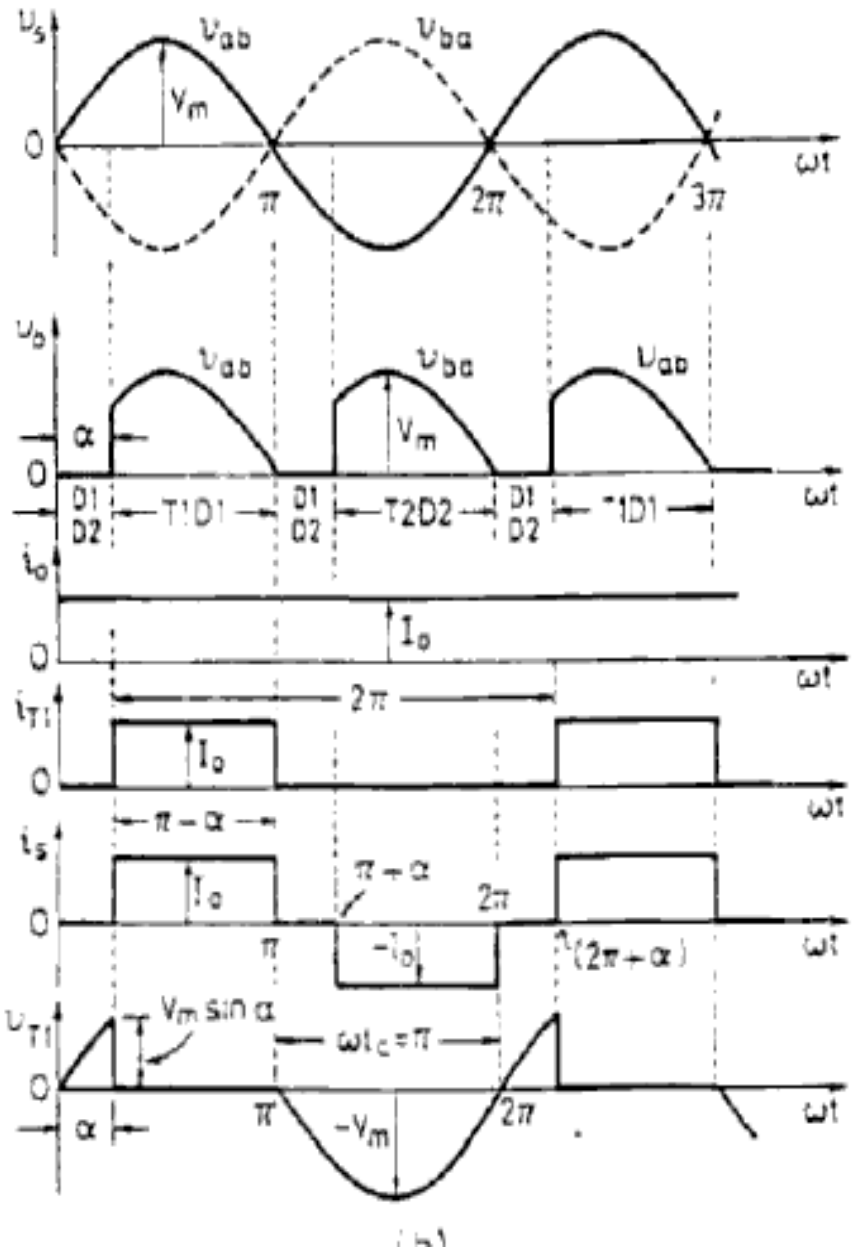
Half controlled converter with RL load (Symmetrical)





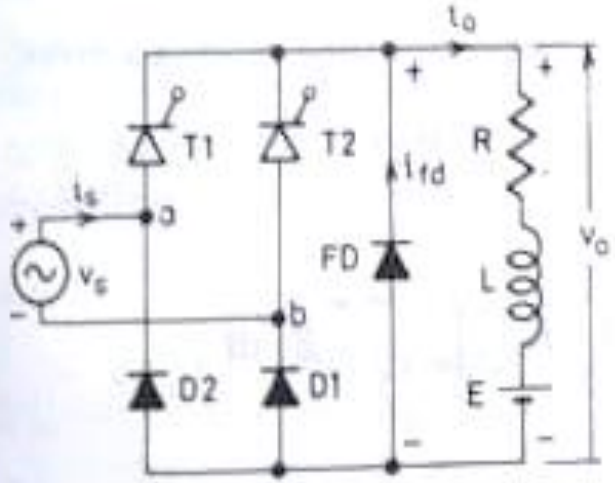
Asymmetrical semi converter with RL load

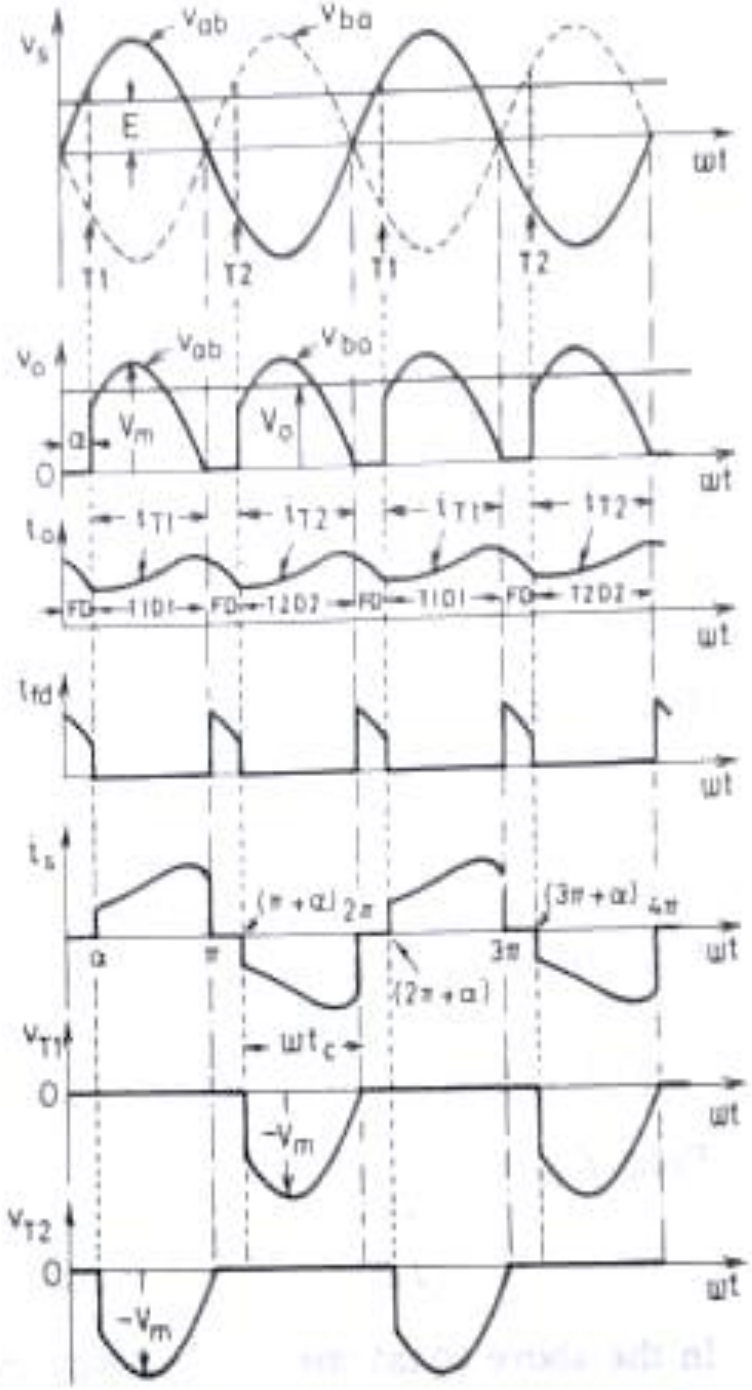




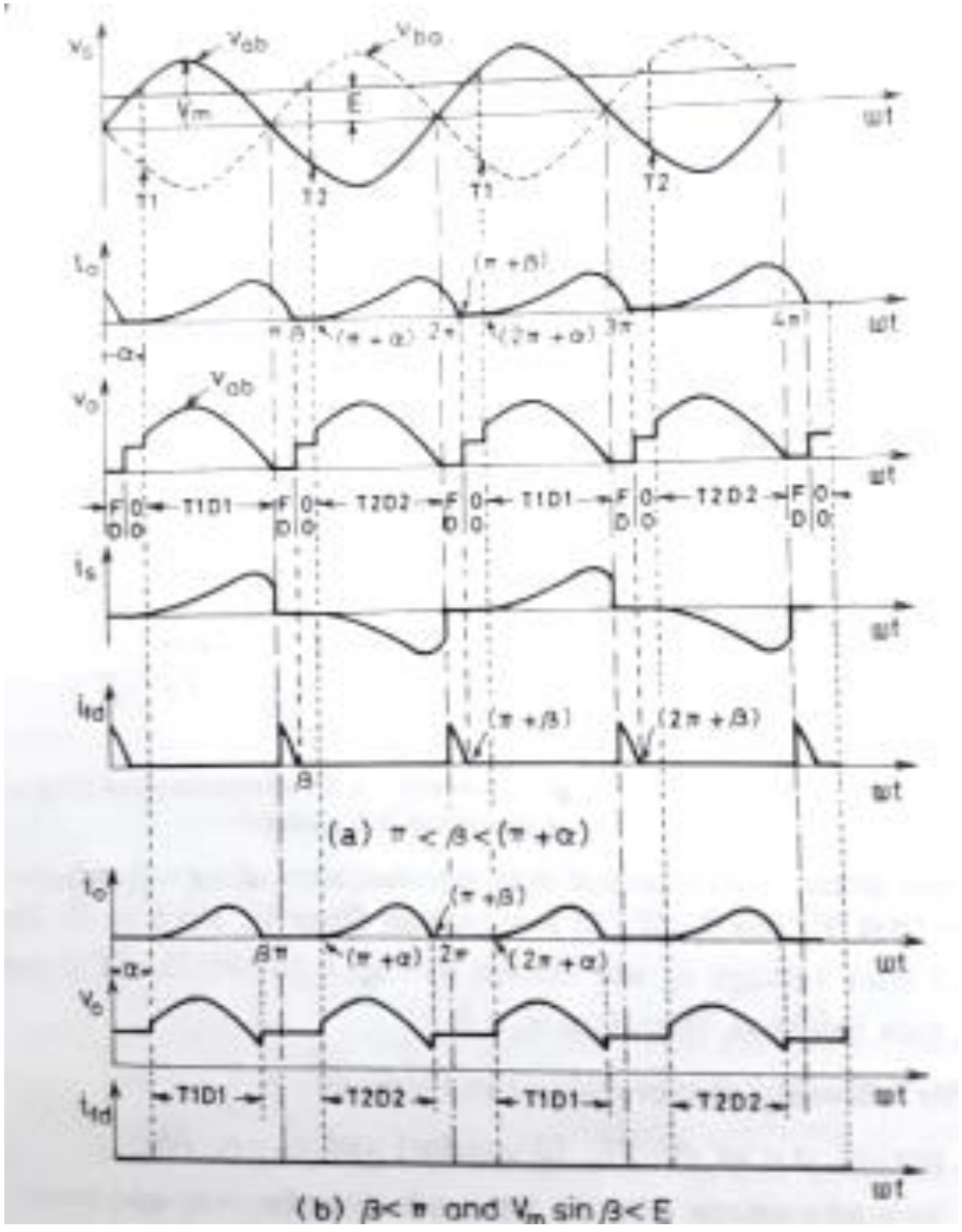
Semiconverter with RLE load

Continuous conduction





Discontinuous conduction

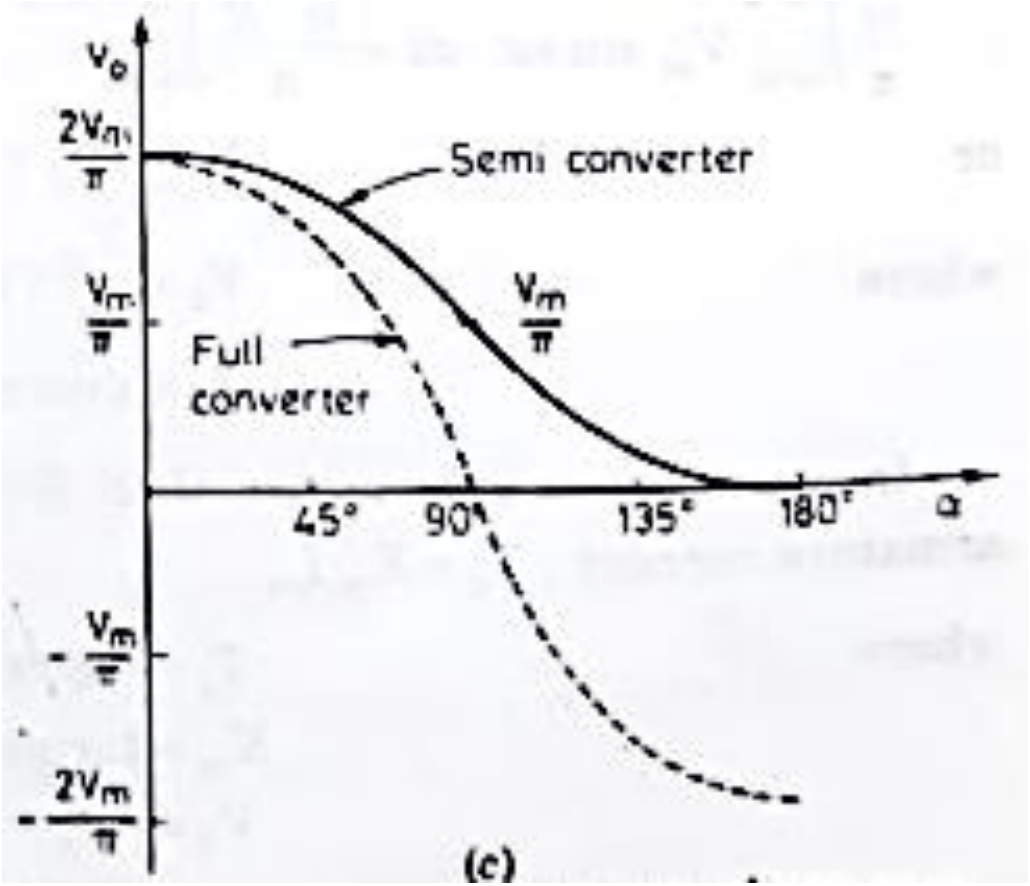


$$V_o = \frac{1}{\pi}\left[\int_{\alpha}^{\pi} V_m \sin \omega t \, . \, d(\omega t) + E(\pi + \alpha - \beta)\right]$$

$$= \frac{V_m}{\pi}(1 + \cos \alpha) + E\left(1 - \frac{\gamma}{\pi}\right)$$

Average load current, $I_o = \frac{V_o}{R} = \frac{V_m}{\pi R}(1 + \cos \alpha) + \frac{E}{R}\left(1 + \frac{\gamma}{\pi}\right)$.

V_o Vs α


V_o
$\frac{2V_m}{\pi}$
Semi converter
$\frac{V_m}{\pi}$
Full converter
45°
90°
135°
180°
α
$-\frac{V_m}{\pi}$
$-\frac{2V_m}{\pi}$
(c)

THD

- The amount of distortion in the voltage or current waveform (here in the input current) is quantified by means of an index called the total harmonic distortion (THD).
- RMS value of distortion component of the current

$$I_{\text{dis}} = [I_s^2 - I_{s1}^2]^{1/2} = \left(\sum_{h \neq 1} I_{sh}^2\right)^{1/2}$$

$$\%\text{THD}_i = 100 \times \frac{I_{\text{dis}}}{I_{s1}}$$

$$= 100 \times \frac{\sqrt{I_s^2 - I_{s1}^2}}{I_{s1}}$$

$$= 100 \times \sqrt{\sum_{h \neq 1} \left(\frac{I_{sh}}{I_{s1}}\right)^2}$$

wer Factor

Average (real) power, $$P = \frac{1}{T_1}\int_0^{T_1} p(t)\,dt = \frac{1}{T_1}\int_0^{T_1} v_s(t)i_s(t)\,dt$$

The integrals of all the
of two different frequency components) are individually zero.

$$P = \frac{1}{T_1}\int_0^{T_1} \sqrt{2}V_s \sin\omega_1 t \cdot \sqrt{2}I_{s1}\sin(\omega_1 t - \phi_1)\,dt = V_s I_{s1}\cos\phi_1$$

$$S = V_s I_s \qquad PF = \frac{P}{S} = \frac{V_s I_{s1}\cos\phi_1}{V_s I_s} = \frac{I_{s1}}{I_s}\cos\phi_1$$

The **displacement power factor (DPF)**, (which is the same as the power factor in linear circuits with sinusoidal voltages and currents) is defined as the cosine of the angle ϕ_1 $\quad \mathrm{DPF} = \cos\phi_1$

Power factor with a non sinusoidal current

$$\text{PF} = \frac{I_{s1}}{I_s}\,\text{DPF}$$

Distortion factor = $\dfrac{I_{s1}}{I_s}$

$$\text{PF} = \frac{1}{\sqrt{1 + \text{THD}_i^2}}$$